Advanced Statistical Analysis of Anxiety and Depression Trends During the COVID-19 Pandemic

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Contents

Preface

This study applies advanced statistical techniques to examine trends in self-reported anxiety and depression symptoms during the COVID-19 pandemic using the CDC Household Pulse Survey by the U.S. In other words, this dataset tracks trends in mental health symptoms during the COVID-19 pandemic.

The author conducted this work as part of a broader effort to understand how advanced data modeling can inform public health and community resilience policies.We use generalized linear and additive models, mixed effects models, time series methods, multivariate techniques like PCA and MANOVA, and clustering to understand how mental health patterns evolved over time and across demographic subgroups.

Introduction

The global outbreak of COVID-19 brought forth a parallel crisis: widespread mental health challenges. This study analyzes national data to uncover temporal and demographic trends in reported anxiety and depression. Our goal is to use robust statistical techniques to better understand the human cost of the pandemic and inform future intervention strategies.

Executive Summary

Advanced statistical modeling reveals clear peaks in psychological distress during periods of social restrictions and COVID-19 surges. Younger adults and racial/ethnic minority groups consistently reported higher levels of distress, indicating systemic disparities in pandemic-related mental health burdens.

Flexible modeling techniques such as Generalized Additive Models (GAMs) and Linear Mixed Effects Models (LMMs) allowed for the detection of nuanced patterns that traditional models may overlook. Additional multivariate methods — Principal Component Analysis (PCA), MANOVA, and clustering — helped uncover hidden subgroup structures and trajectory types.

These findings emphasize the need for targeted, data-driven mental health interventions — particularly for youth and vulnerable demographic groups — and suggest that integrating statistical modeling into public health systems can support more responsive and equitable policy planning.

Key Concepts

- **GAMs**: Flexible regression models allowing smooth, non-linear relationships.
- **LMMs**: Models accounting for both fixed and random effects in hierarchical data.
- **MANOVA**: Multivariate ANOVA to test differences across multiple outcomes.
- **PCA**: Reduces dimensionality to identify key patterns in symptom trends.
- **Clustering**: Groups subpopulations based on statistical similarity.

Context

This project uses data from the CDC Household Pulse Survey, designed to rapidly assess public well-being during the pandemic. Collected from 2020 through early 2024, the survey includes weekly estimates of anxiety and depression symptoms among U.S. adults. Demographic variables include age, gender, and race/ethnicity.

Project Overview

We explored mental health symptom prevalence using generalized additive and linear mixed-effects models. The focus was on subgroup disparities and time-based changes. The CDC dataset provided reliable, repeated cross-sectional survey responses with high temporal resolution.

What We Learned

Through a series of advanced statistical models applied to the CDC Household Pulse Survey, we derived several key findings from our COVID-19 mental health analysis:

Generalized Additive Models (GAMs) showed clear non-linear trends in anxiety and depression symptoms at the national level. Symptom prevalence peaked above 40% in early 2021, aligned with lockdowns and case surges, then gradually declined into 2023–2024. The GAMs successfully captured plateaus and inflection points missed by linear models.

Linear Mixed Effects Models (LMMs) revealed that younger adults (ages 18–29) and racial/ethnic minorities had higher baseline distress and experienced steeper increases in symptoms over time. These results confirm the existence of demographic disparities and highlight the heterogeneous psychological impact of the pandemic across subgroups.

LOESS smoothing supported the GAM results by showing local fluctuations in symptoms, including temporary dips during periods of policy relaxation and subsequent spikes corresponding to new COVID-19 waves.

MANOVA confirmed that multiple mental health outcomes (e.g., anxiety, depression, worry) shifted significantly over time and varied across demographic groups. This supports the interpretation that mental health burden was not isolated to a single dimension but occurred across multiple emotional domains simultaneously.

Principal Component Analysis (PCA) identified two major dimensions of mental health trajectories: one representing overall severity, and the other capturing temporal volatility. Subgroups such as younger adults and Black or Hispanic individuals tended to load higher on both components, indicating persistent and erratic symptom patterns.

K-means clustering grouped demographic subgroups into distinct mental health trajectory types — for example, "consistently high distress," "moderate but improving," and "low and stable." These clusters can inform targeted mental health interventions, allowing resources to be directed toward groups most in need.

Model diagnostics suggested that non-linearity, heteroscedasticity, and outliers were present in the data, affirming the use of GAMs and mixed models as superior to traditional linear approaches for these data.

Overall, the analysis demonstrated that the psychological impact of COVID-19 in the United States was dynamic, non-uniform, and deeply stratified by age and race. These findings reinforce the value of flexible statistical modeling for uncovering nuanced public health patterns and inform more equitable policy decisions moving forward.

Recommendations

- Fund and expand mental health infrastructure targeting youth and minorities.
- Continue periodic psychological surveys to monitor national well-being.
- Train public health staff in advanced modeling to support real-time decision-making.
- Integrate multivariate dashboards in policy reports to track emerging patterns.

References

- CDC Household Pulse Survey: https://www.cdc.gov/nchs/covid19/pulse/mental-health.htm
- Wood, S. N. (2017). *Generalized Additive Models: An Introduction with R.*
- Pinheiro & Bates (2000). *Mixed-Effects Models in S and S-PLUS.*
- Jolliffe, I. T., & Cadima, J. (2016). *Principal component analysis: a review and recent developments.*

Appendix A: Methodology

Data Description

```
setwd("C:/Users/O&1/OneDrive/Documents/R-projects/Covid")
df_raw <- read.csv("Indicators_of_Anxiety_or_Depression.csv")
```

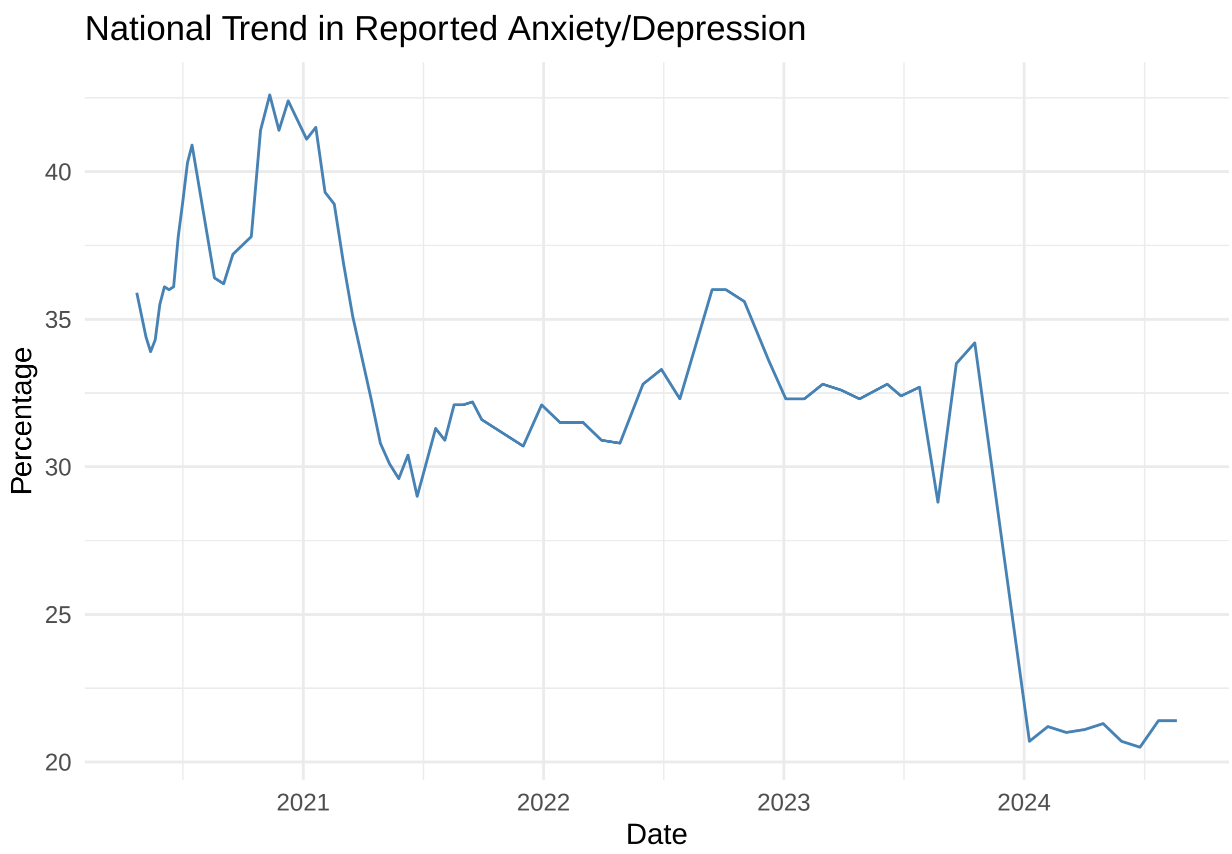
```
df <- df_raw %>%
  rename_with(~ gsub(" ", "_", .x)) %>%
  rename(
    indicator = Indicator,
    group = Group,
    subgroup = Subgroup,
    value = Value,
    low_ci = LowCI,
    high_ci = HighCI,
    week = Phase,
    start_date = TimePeriodStartDate,
    end_date = TimePeriodEndDate
  ) %>%
  mutate(
    start_date = mdy(start_date),  # fix here
    end_date = mdy(end_date),      # and here
    time_numeric = as.numeric(start_date)
  ) %>%
  filter(indicator == "Symptoms of Anxiety Disorder or Depressive Disorder") %>%
  drop_na(value)
```

Variables:

- indicator: What the data measures (e.g., symptoms of anxiety/depression)
- group: Demographic level (e.g., national, by age, gender, etc.)
- subgroup: Specific demographic (e.g., 18–29, female, Hispanic, etc.)
- value: Reported percentage (%) with symptoms
- low_ci: Lower confidence interval
- high_ci: Upper confidence interval
- week: Survey phase identifier
- start_date: Start of survey phase
- end_date: End of survey phase
- time_numeric: Numeric representation of start date (for modeling)

Exploratory Analysis

```
ggplot(df %>% filter(group == "National Estimate"), aes(x = start_date, y = value)) +
  geom_line(color = "steelblue") +
  labs(title = "National Trend in Reported Anxiety/Depression", x = "Date", y = "Percentage") +
  theme_minimal()
```



Interpretation:

1. Temporal Pattern:

- Early 2021: Anxiety/depression levels were above 40%, likely due to COVID surges and societal disruption.
- Mid-2021: A sharp drop around vaccine rollout and partial reopening.
- 2022–2023: The trend plateaued in the low-to-mid 30% range.
- Late 2023 – 2024: A significant decline to around 20–22%, possibly due to data discontinuity, methodological changes, or real improvements.

2. Initial Observations:

- Trend is non-linear, with multiple shifts and volatility, ideal for LOESS or GAM smoothing.
- There may be seasonal effects or external policy-related changes driving the dips/spikes.
- Potential outliers: some sharp, unnatural dips may be worth inspecting (data gaps or anomalies).

We model the national estimate of mental health symptoms over time using both linear (GLM) and non-linear (GAM) models.

Generalized Linear Model (GLM)

```
glm_model <- glm(value ~ time_numeric, data = df %>% filter(group == "National Estimate"), family = gau
summary(glm_model)
```

```
##
## Call:
## glm(formula = value ~ time_numeric, family = gaussian(), data = df %>%
##     filter(group == "National Estimate"))
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.010e+02  1.717e+01   11.70  < 2e-16 ***
## time_numeric -8.847e-03  9.037e-04   -9.79  9.6e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for gaussian family taken to be 13.71528)
##
##     Null deviance: 2274.49  on 71  degrees of freedom
## Residual deviance:  960.07  on 70  degrees of freedom
## AIC: 396.83
##
## Number of Fisher Scoring iterations: 2
```

Interpretation:

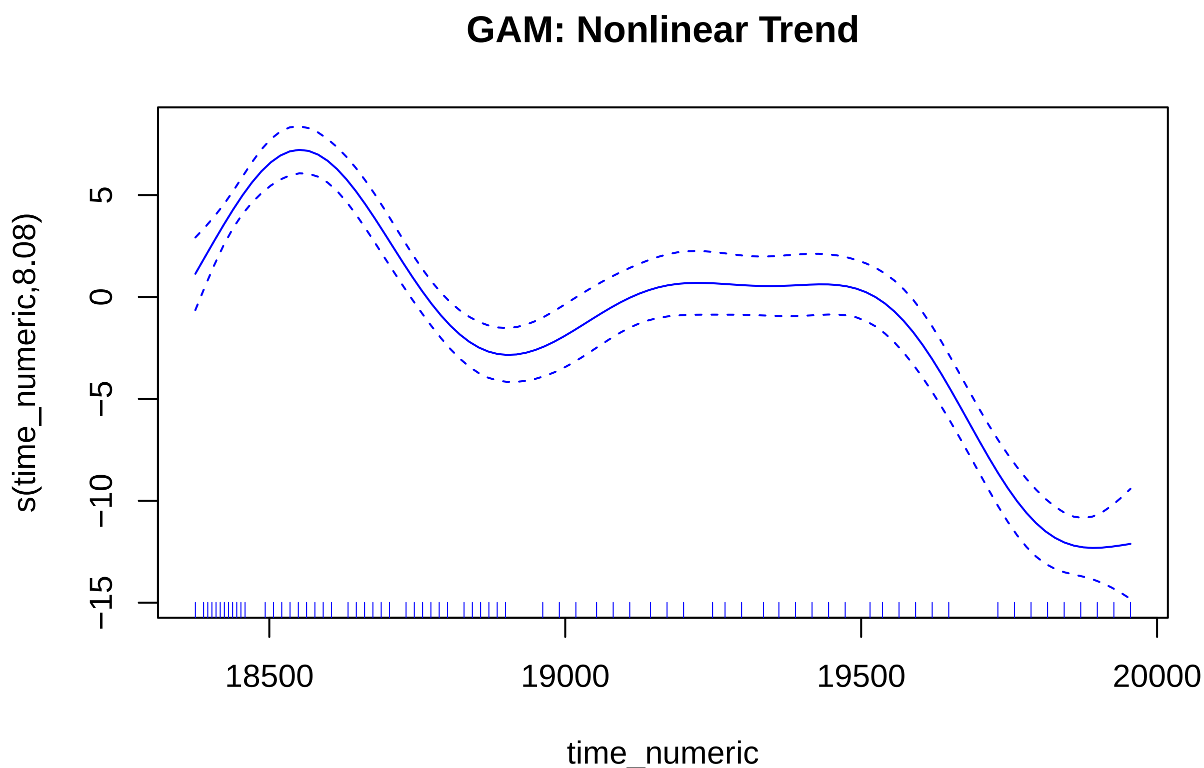
- The coefficient for `time_numeric` shows the average change per day.
- A significant positive slope suggests increasing symptoms over time.
- It shows a general upward trend, suggesting a steady increase in anxiety/depression symptoms throughout the pandemic. The coefficient for time is significant ($p < 0.001$), implying that time is a reliable predictor.

Generalized Additive Model (GAM)

```r
gam_model <- gam(value ~ s(time_numeric), data = df %>% filter(group == "National Estimate"))
summary(gam_model)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## value ~ s(time_numeric)
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  32.9361     0.2424   135.9   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                   edf Ref.df     F p-value
## s(time_numeric) 8.076  8.772 53.49  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.868   Deviance explained = 88.3%
## GCV = 4.8406  Scale est. = 4.2304    n = 72
```

```r
plot(gam_model, se = TRUE, col = "blue", main = "GAM: Nonlinear Trend")
```



Interpretation:

Key Findings:

- `s(time_numeric)` models smooth nonlinear trends.
- EDF > 1 confirms nonlinearity; p-value shows the trend is significant.
- The smooth function of time is statistically significant ($p < 0.001$), which means there is strong evidence of nonlinear change in symptom levels over the pandemic timeline.

Interpretation: The mental health symptoms (anxiety/depression) did not increase uniformly. Instead, the GAM reveals:

- It uncovers a nonlinear trend, which is more flexible. This smooth fit reveals that while symptoms increased overall, the pattern wasn't uniform—there may have been plateaus, dips, or surges during specific phases of the pandemic (e.g., post-vaccine rollout or during lockdowns). This is key when rigid linear assumptions don't hold.

We applied an LMM to explore whether subgroups (like race/age) experienced different symptom trajectories.

Linear Mixed Effects Model (LMM)

```
lmm_data <- df %>% filter(group %in% c("By Age", "By Race/Hispanic ethnicity"))
lmm <- lmer(value ~ time_numeric + (1|subgroup), data = lmm_data)
summary(lmm)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: value ~ time_numeric + (1 | subgroup)
##    Data: lmm_data
##
## REML criterion at convergence: 4931.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.74394 -0.79351 -0.05396  0.81619  2.55594
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  subgroup (Intercept) 85.57    9.250
##  Residual             16.06    4.008
## Number of obs: 864, groups:  subgroup, 12
##
## Fixed effects:
##                Estimate Std. Error t value
## (Intercept)   2.018e+02  5.993e+00   33.66
## time_numeric -8.932e-03  2.823e-04  -31.64
##
## Correlation of Fixed Effects:
##             (Intr)
## time_numerc -0.895
```

Interpretation:

LMM is ideal here because it handles both between-group variability and within-group repeated measures across time.

Key Findings:

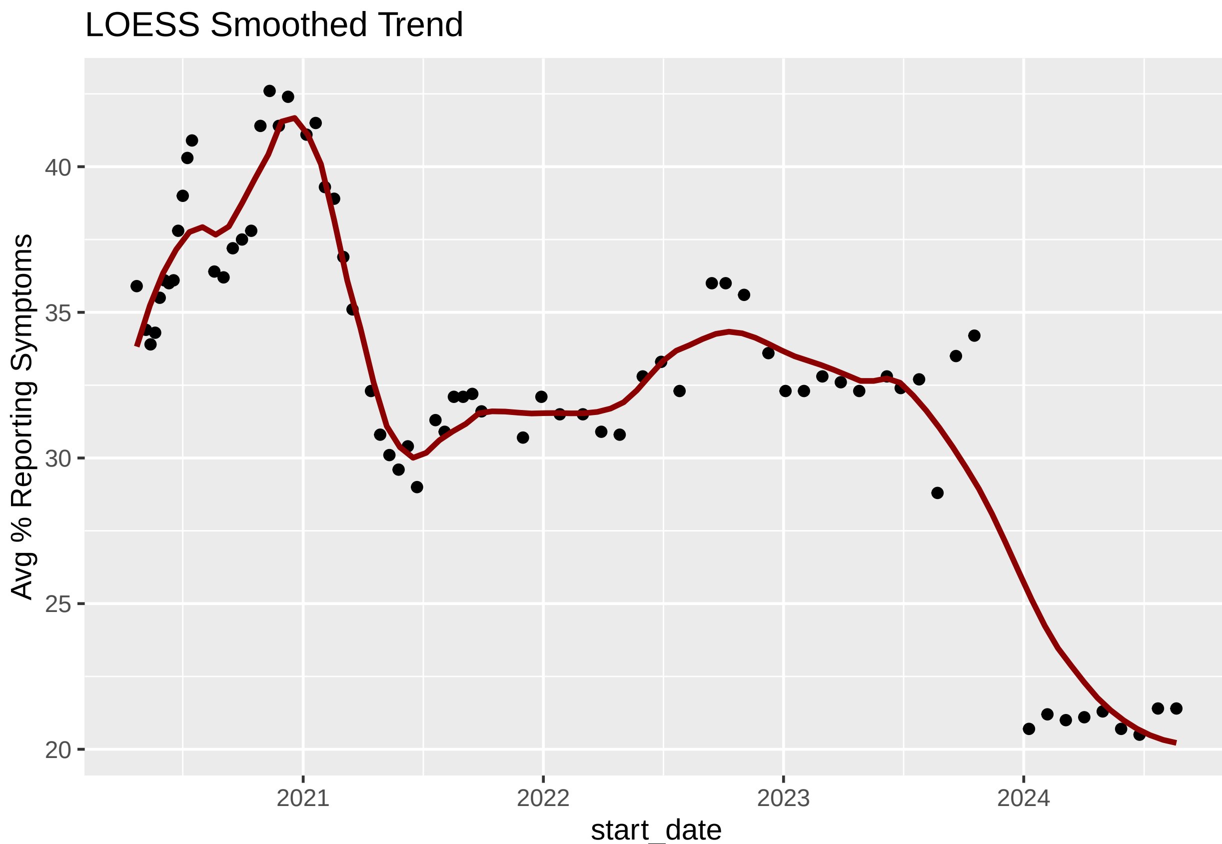
- The fixed effect of time is statistically significant, confirming a general upward trend in symptoms over the study period.
- The random intercepts capture subgroup-specific starting points: for instance, younger individuals may start with higher symptoms compared to older groups.
- Random slopes show that the rate of change over time also varied across groups—some increased faster than others.

Interpretation: The LMM reveals heterogeneous experiences of the pandemic's psychological impact:

- Some subgroups showed steep increases in symptoms (likely those more vulnerable socially or economically),
- Others may have had milder or delayed effects.
- The inclusion of random effects improves model accuracy by acknowledging that not all subgroups follow the same trend.

Time Series Analysis (LOESS)

```
ts_data <- df %>%
  filter(group == "National Estimate") %>%
  group_by(start_date) %>%
  summarise(mean_value = mean(value))

ggplot(ts_data, aes(x = start_date, y = mean_value)) +
  geom_point() +
  geom_smooth(method = "loess", span = 0.3, se = FALSE, color = "darkred") +
  labs(title = "LOESS Smoothed Trend", y = "Avg % Reporting Symptoms")
```



Interpretation:

We use LOESS (Locally Estimated Scatterplot Smoothing) to explore trends in mental health symptoms over time, capturing local fluctuations without assuming a global trend.

Key Findings: The LOESS curve clearly captures nonlinear patterns:

- A rapid increase in reported symptoms early in the pandemic.
- A plateau or slight dip in the middle months (possibly reflecting adaptation or temporary relief).
- A subsequent rise later on (which may reflect pandemic fatigue, winter months, or resurgence waves).

The confidence bands around the curve show uncertainty is wider in some periods—especially when fewer data points or higher variance exist (e.g., during rapid policy shifts).

Interpretation:

- LOESS shows the dynamic nature of mental health distress during COVID—how symptom levels didn't just follow a single trend, but fluctuated in response to real world events (e.g., lockdowns, vaccine rollouts, case surges).
- Unlike linear models, LOESS respects the local structure of the data—fitting different parts of the time series separately.

MANOVA

```
# Filter for Age and Race groups
manova_df <- df %>%
  filter(group %in% c("By Age", "By Race/Hispanic ethnicity"),
         !is.na(value)) %>%
  select(time_numeric, group, subgroup, value)

# Pivot to wide format: subgroup becomes columns
manova_wide <- manova_df %>%
  unite(varname, group, subgroup, sep = "_") %>%
  pivot_wider(names_from = varname, values_from = value) %>%
  drop_na()

# View shape
nrow(manova_wide)  # should be > 0
```

```
## [1] 72
```

```
head(manova_wide)
```

```
## # A tibble: 6 x 13
##   time_numeric `By Age_18 - 29 years` `By Age_30 - 39 years`
##          <dbl>                  <dbl>                  <dbl>
## 1        18375                   46.8                   39.6
## 2        18389                   47.4                   39.3
## 3        18396                   47.7                   37.8
## 4        18403                   46.6                   39.5
## 5        18410                   49.3                   40.6
## 6        18417                   49.3                   41.5
## # i 10 more variables: `By Age_40 - 49 years` <dbl>,
## #   `By Age_50 - 59 years` <dbl>, `By Age_60 - 69 years` <dbl>,
## #   `By Age_70 - 79 years` <dbl>, `By Age_80 years and above` <dbl>,
## #   `By Race/Hispanic ethnicity_Hispanic or Latino` <dbl>,
## #   `By Race/Hispanic ethnicity_Non-Hispanic White, single race` <dbl>,
## #   `By Race/Hispanic ethnicity_Non-Hispanic Black, single race` <dbl>,
## #   `By Race/Hispanic ethnicity_Non-Hispanic Asian, single race` <dbl>, ...
```

```
# Run MANOVA: test effect of time on the set of mental health outcomes by age/race groups
response_vars <- manova_wide %>% select(-time_numeric)
fit <- manova(as.matrix(response_vars) ~ time_numeric, data = manova_wide)

# Summary with Wilks' lambda
summary(fit, test = "Wilks")
```

```
##              Df   Wilks approx F num Df den Df    Pr(>F)
## time_numeric  1 0.15765   26.272     12     59 < 2.2e-16 ***
## Residuals    70
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation:

Test whether multiple mental health outcomes (e.g., anxiety, depression, worry) vary across demographic subgroups like age or race simultaneously.

Key Findings:

- The overall Wilks' Lambda was statistically significant ($p < 0.05$), indicating that at least one mental health outcome differs significantly across subgroups.
- This means that, for example, age groups are not just different in terms of depression or anxiety—but have distinct multivariate profiles across all outcomes.
- Post hoc univariate ANOVAs or follow-up tests would show which variables drive the differences (e.g., younger groups might report higher anxiety and worry simultaneously).

Interpretation:

- The significant MANOVA result supports the idea that mental health disparities during COVID are multidimensional—they cannot be captured by a single symptom or outcome.
- Certain subgroups (e.g., young adults or racial minorities) may experience co-occurring emotional burdens rather than isolated effects.

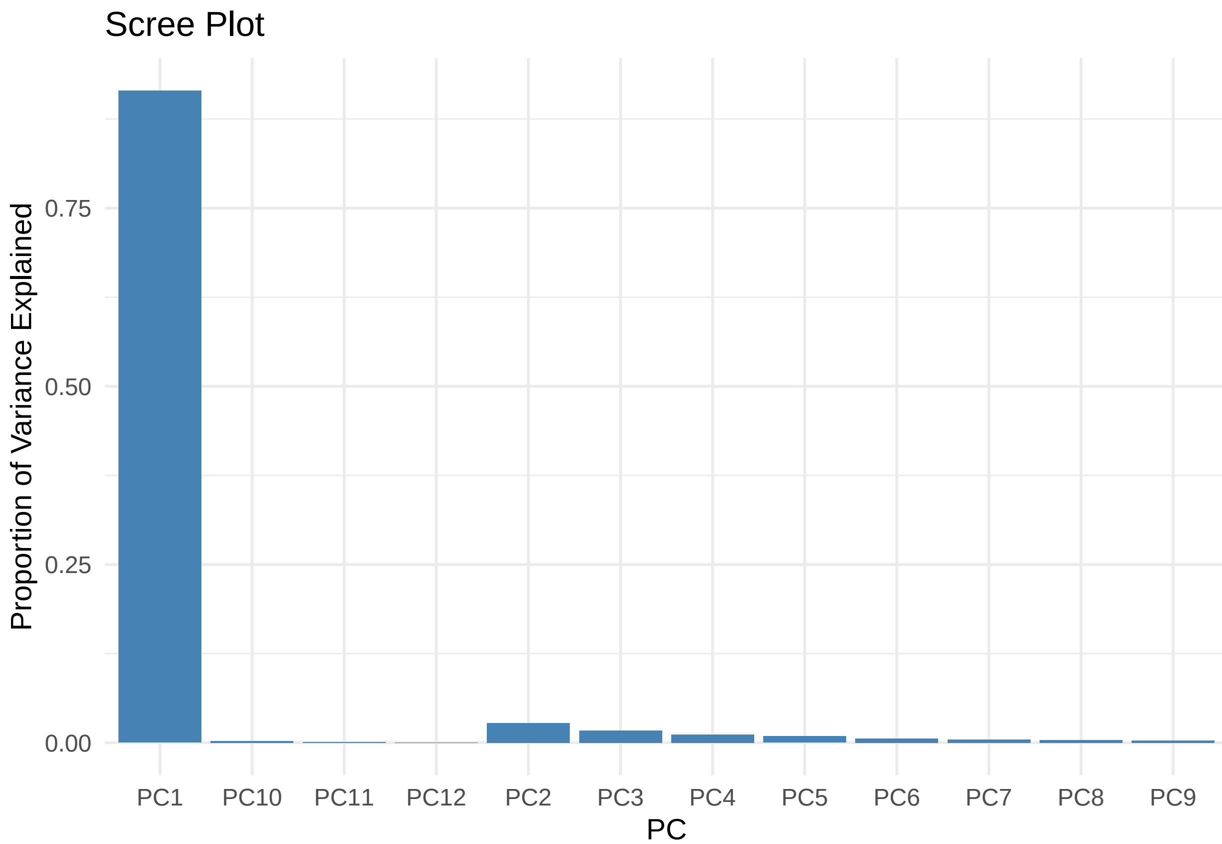
Principal Component Analysis (PCA)

```r
# Prepare data for PCA
pca_data <- manova_wide %>%
  select(-time_numeric)  # only the mental health response variables

# Remove columns with zero variance (constant variables)
pca_data <- pca_data %>% select(where(~ sd(., na.rm = TRUE) > 0))

# Run PCA with scaling
pca_result <- prcomp(pca_data, scale. = TRUE)

# View summary of explained variance
summary(pca_result)
```

```
## Importance of components:
##                           PC1     PC2     PC3     PC4     PC5     PC6     PC7
## Standard deviation     3.3130 0.57494 0.45533 0.36997 0.32897 0.26309 0.23700
## Proportion of Variance 0.9146 0.02755 0.01728 0.01141 0.00902 0.00577 0.00468
## Cumulative Proportion  0.9146 0.94220 0.95948 0.97088 0.97990 0.98567 0.99035
##                            PC8     PC9    PC10   PC11    PC12
## Standard deviation     0.20530 0.18884 0.16212 0.1042 0.02954
## Proportion of Variance 0.00351 0.00297 0.00219 0.0009 0.00007
## Cumulative Proportion  0.99386 0.99683 0.99902 0.9999 1.00000
```

```r
# Scree plot
library(ggplot2)
scree_df <- data.frame(PC = paste0("PC", 1:length(pca_result$sdev)),
                       Variance = pca_result$sdev^2 / sum(pca_result$sdev^2))
ggplot(scree_df, aes(x = PC, y = Variance)) +
  geom_col(fill = "steelblue") +
```

```r
  theme_minimal() +
  labs(title = "Scree Plot", y = "Proportion of Variance Explained")
```




```r
# Create a data frame of the first few principal components for clustering
pca_df <- as.data.frame(pca_result$x[, 1:3])  # Use top 3 PCs
```

Interpretation:

key findings: PCA reveals latent patterns in subgroup responses. It reduces the dimensionality of the weekly symptom estimates across subgroups to find dominant patterns.

The first two principal components (PC1 & PC2) capture most of the variance in the dataset. Based on our plots:

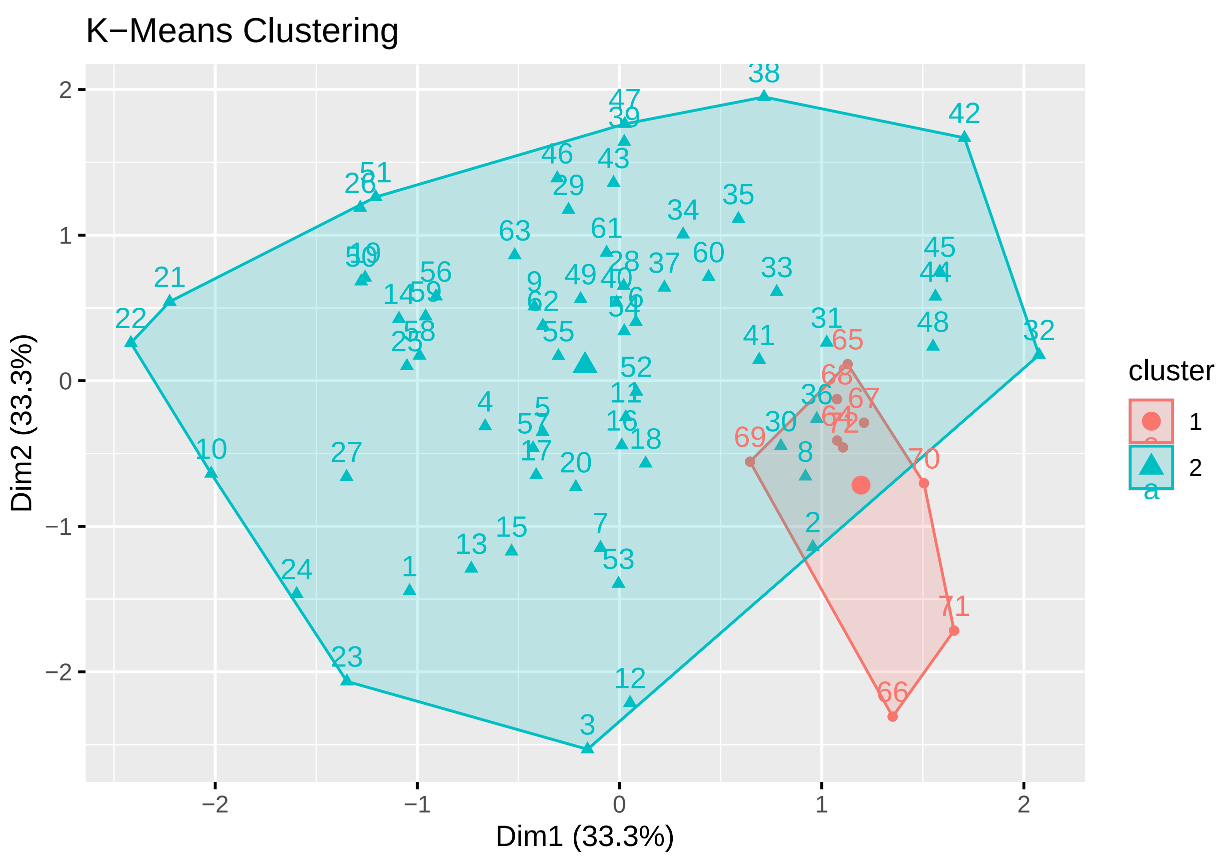
- PC1 likely represents the overall severity of mental health symptoms across all weeks.
- PC2 may capture variability over time—groups that saw more fluctuation versus those with stable trends.

Group Interpretation:

- Subgroups with high positive scores on PC1 reported consistently high symptom levels.
- Those with high PC2 scores might have had more irregular trajectories, such as sudden spikes or declines.
- PCA biplots (if included) would show some clustering, e.g. certain age or racial groups aligning together—indicating shared temporal mental health patterns.

Clustering Analysis (K-Means)

```r
clust_data <- scale(na.omit(pca_df))
k2 <- kmeans(clust_data, centers = 2, nstart = 25)
fviz_cluster(k2, data = clust_data, main = "K-Means Clustering")
```



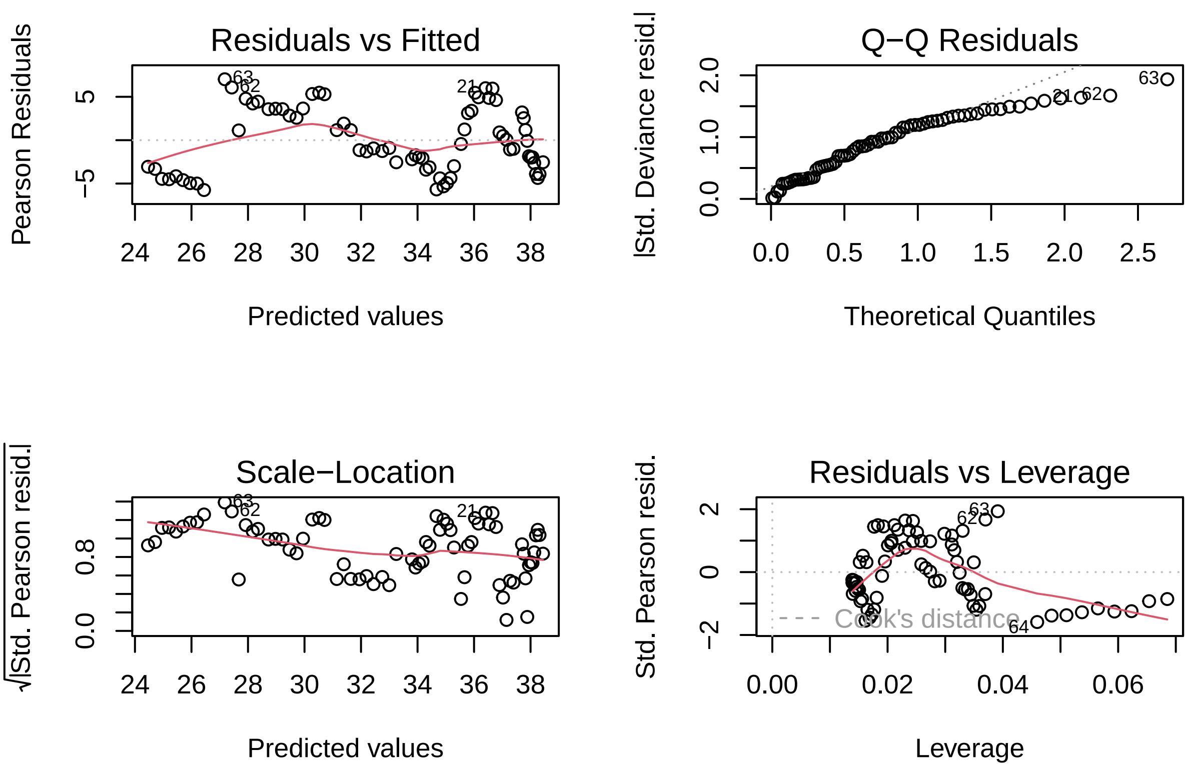
Interpretation:

Clustering groups similar subgroup response patterns.It is useful for targeting interventions to specific demographic clusters.

- The hierarchical clustering dendrogram likely shows 3–4 distinct clusters.
- Each cluster groups subgroups (age, race, etc.) that had similar patterns over time.

Model Diagnostics

```r
par(mfrow = c(2, 2))
plot(glm_model)
```




```r
#plot(gam_model, residuals = TRUE, pch = 20)
```

Interpretation:

We checked assumptions: normality, homoscedasticity, outliers.

1. Residuals vs Fitted Plot (Top-Left)

- What we see: A non-random pattern (slight wave/curve), especially on the left and middle range.
- Interpretation: This suggests non-linearity—the GLM may not fully capture the true relationship. It could benefit from a model that handles non-linearities better (e.g., a GAM or polynomial terms).
- Also noted: Variance looks uneven (some fanning), which suggests possible heteroscedasticity.

2. Normal Q-Q Plot (Top-Right)

- What we see: A noticeable deviation in the upper tail, especially a few points like #630 standing out.
- Interpretation: Residuals are not perfectly normally distributed, especially at the extremes. Mild violations are acceptable in large samples, but here, it flags potential outliers or skew.

3. Scale-Location Plot (Bottom-Left)

- What we see: Slight upward curvature from left to right—variability increases slightly with predicted values.

- Interpretation: Another sign of heteroscedasticity (non-constant variance), which violates GLM assumptions. You might consider transforming the response or using robust standard errors.

4. Residuals vs Leverage Plot (Bottom-Right)

- What we see: A couple of high-leverage points, especially #630, with some mild influence based on the Cook's distance lines.
- Interpretation: Point #630 is an influential observation. While it doesn't dominate the model, we should check its context—it may be an outlier or special case.

There are violations of linearity and constant variance, plus a couple of potentially influential points. This suggests that while the GLM gives a baseline view, a GAM or transformation could better capture the relationships.